Automated Handwritten Essay Evaluation in Moodle: Leveraging Google Vision OCR and Mistral 7b

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Abstract

Traditional grading methods are often time-consuming and subjective, increasing the difficulties in maintaining academic integrity against the background of easy access to online resources. Even as pre-written and AI-generated content become even more available, handwritten essays offer one of the most viable ways of stimulating real learning and original thought among students. This study assessed the accuracy and efficiency of AI-powered grading to improve education amid these challenges. More precisely, the paper investigated the possibility of automatically grading the handwritten open-ended reflection essays of students in the "Living in the IT Era" course by leveraging AI. Through Optical Character Recognition (OCR), together with a fine-tuned and trained Large Language Model (LLM) Mistral 7b, the system replicated human-grading decisions in evaluating essays comprehensively. The predicted score and human-graded score were compared in evaluating the system. Analysis using BERT Score revealed a high degree of correlation between the two, with consistent precision (88.41%), recall (83.33%), and F1-score (85.78%). External user testing also revealed a positive perception: the system is perceived as user-friendly (usability: 4.28), generally understandable (comprehensibility: 3.68), and with functionalities relevant to educators' needs (relevance: 4.0). This study adds to the expanding body of research on AI in education and lays the groundwork for future investigations in this area.

CCS Concepts

• **Computing methodologies**; • **Artificial intelligence**; • **Computer vision;**;

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Handwritten Essay Grading, Essay Evaluation, Moodle, Plugin, Online Education, Learning Management Systems

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1 Introduction

Although essays happen to be one of the strong measures in determining a student's grasp toward a given subject, the digital age has introduced a new: copying from online content. The ease of access to pre-written essays online and generation of content by AI could create shortcuts that undermine true learning and academic integrity. There is a stark advantage for handwritten essays in this context. They require students to actively process information, formulate arguments, and organize their thoughts. This active learning process promotes a much stronger grasp of the subject matter, more original thought, and clearer understanding of context. Additionally, the act of handwriting forces a student to absorb what they are writing.

This study explored the viability and efficacy of employing AI tools to automate the evaluation of handwritten open-ended student essays within the "Living in the IT Era" course. The potential benefits and limitations were investigated, focused on accuracy, efficiency, and ability to provide immediate score and feedback to students. To achieve this, a multifaceted approach was adopted, combining Optical Character Recognition (OCR) and a Large Language Model (LLM), enabling the evaluation of diverse aspects of writing by leveraging machine learning techniques.

The system was integrated with the Learning Management System – Moodle – which was being used by the university, to assess its potential improvements in grading efficiency and effectiveness for educators. The research focused on the "Living in the IT Era Course" and determined how effective an AI-based grading system would be in this particular context of education, which in the future may pave the way for a broader application.

2 Related Works

One major advantage of the usage of Artificial Intelligence generally comes with efficiency and speed of doing its task. Lately, there has been an increased usage of Artificial Intelligence in education, mostly in the area of automating student assessments.

2.1 Optical Character Recognition and Handwritten Optical Character Recognition

Optical Character Recognition is a science that enables us to translate printed documents or images into an editable data format [1]. OCR is one of the main branches of Artificial Intelligence that inputs text in machine-coded form. However, traditional OCR is primarily oriented toward printed text; recently, it has been modified to include handwritten text recognition in its fold to give it another synonym—"Handwritten OCR". OCR, being a technology applied to handwritten documents, which are very common in various fields like education, archival preservation, historical research, and so on, identifies and digitizes the handwritten content to make many documents more accessible.

2.2 Handwriting Recognition

Handwriting recognition, also known as Handwritten Text Recognition (HTR), is a branch of artificial intelligence and pattern recognition that focuses on converting handwritten text into machine-readable characters. The landscape of handwriting recognition has been significantly enriched by extensive research efforts in the areas of feature extraction, deep learning architectures, and dataset development. The paper [2] explored the application of neuroevolution to the automatic design of convolutional neural networks (CNNs). It developed two novel solutions based on genetic algorithms and grammatical evolution and evaluated them using the MNIST dataset for handwritten digit recognition, achieving highly competitive results.

A modified topology was presented for Long Short-Term Memory Recurrent Neural Networks (LSTM-RNNs) [3] which controlled the shape of the squashing functions within gating units. It also introduced an efficient training framework based on mini-batch training at the sequence level, combined with a sequence chunking approach. The paper was evaluated on publicly available datasets and outperformed state-of-the-art recognition results, achieving speedups of more than 3x.

2.3 AI in Education

The AIEd or Artificial Intelligence in Education community is also increasingly turning its attention to the ways that AI systems impact online education [4]. Recently artificial intelligence integrated into educational assessment has brought about huge change in the way students are assessed and fed back. In addition, AI effectively supports online learning and teaching, including personalization in learning for students, automation of routine tasks from instructors, and powering adaptive assessments [5].

2.4 Automated Essay Scoring

Recently, developments in the field of automated evaluation techniques for handwritten answer scripts and essays have turned out to be quite accurate, involving deep learning architectures. A deep learning architecture, which combined Convolutional Neural Network and Bidirectional Long Short-Term Memory components for automating the evaluation of handwritten answer scripts, could show an approximate 80% accuracy rate from testing [6]. This performance underlined that developing models for lengthier texts—from 200 to 250 words—incorporating figures and equations was relatively challenging and asked for deeper analysis and research.

In this study, Optical Handwriting Recognition and Automated Essay Scoring technologies were used to automate a grading system for handwritten essays [7]. The system could handle image-based handwritten essays and could provide grading between 0 and 5. The results showed a Quadratic Weighted Kappa score of 0.88, which underscores the power of OHR systems in the context of AES. The work used the Hewlett Foundation AES dataset to help the integration of OHR and AES systems for automated grading of handwritten essays.

2.5 Automated Writing Evaluation Systems

Automated Writing Evaluation systems provides formative feedback which aids students in the refinement of essays. At the moment, this system depends on non-neural attributes for the provision of insight into specific aspects of essay quality. That is, the AWE systems combine computer science effort with natural language processing and latent semantic analysis [8]. Advanced NLP and Machine Learning empower these engines to study writing across lexical, syntactic, semantic, and discourse dimensions. Such systems can emit scores not only regarding the quality of the writing but even provide feedback on how well one is writing [8].

2.6 Natural Language Processing

Natural Language Processing is actually an area of artificial intelligence and computer science that deals with human–computer interaction [9]. NLP is oriented in the ability of computers to process, interpret, and produce human language products in a way that is effective and efficient in terms of computation. Automated essay grading is but one of the many applications in NLP [10]. In this vein, one recent study demonstrates the increase in sophistication of NLP models and an improvement in the accuracy and efficiency of these systems [11]. These models are able to understand even complex structures in language, idioms, and, moreover, cultural references in essays that they check. It has resulted in better-contextured, more accurate essay grading, lower workloads of human graders, and quicker, more consistent feedback to students.

2.7 Large Language Model

Large Language Models or LLMs are a fast-growing subfield of Natural Language Processing. They are also now trained over large bodies of text. They are trained on a huge pool of text to achieve human-level proficiency in doing a lot of different tasks that include generation, translation of text from one language into another, and text classification [12]. Their effectiveness over a variety of NLP applications is extensively documented [13].

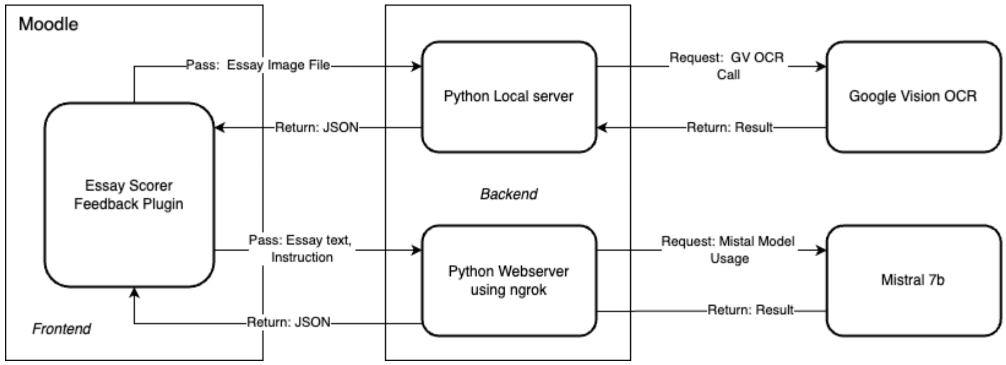

Figure 1: Block Diagram of the Moodle Essay Scorer Feedback Plugin.

2.8 Mistral AI

Mistral AI is one of the newest open-source LLM projects that exists today and it is gaining a lot of attention as well [14]. Being developed by researchers from some of the top AI firms, Mistral provides competitive performance in reasoning and overall capabilities, matching leading LLMs of today [15]. This open-source approach leads to collaboration within the developer community and expands the potential use of LLM technology.

Mistral AI offers a 7.3 billion parameter language model, it is called Mistral 7B, close to the performance of other LLMs in reasoning and overall capabilities. The model is designed for easy deployment and fine-tuning on specific tasks while posting competitive performance on various benchmarks and tasks [14]. This model was able outperform the some of the well-known LLMS, like the 13 billion parameter Llama 2 model, it outperformed it on all tasks and as well as the 34 billion parameter Llama 1 on many benchmarks124. Mistral 7B performs almost as well as the CodeLlama 7B on coding and still significantly outperforms it on English tasks.

2.9 Humans and AES Engines

Two studies conducted to evaluate the performances of Automated Essay Scoring engines against human raters showed that the AES engines outperformed human raters on five of the seven measures and closely approximated human rater performance on the other two, confirming the potential of AES for high-stakes writing assessments as a useful tool [16].

3 Methodology

The study utilized a multi-step procedure in developing the automated essay evaluation system. Figure 1 shows the block diagram of the Moodle Essay Scorer Feedback Plugin. It elaborates on the components and processes of the plugin and how the data is passed from Moodle through to the backend servers. This figure demonstrates that the data will go through various processing stages.

3.1 Data Collection

Essays were gathered from submissions of student activities from the course titled "IT031 - Living in the IT Era." Authored solely by second-year IT students in the academic year 2022-2023, the essays were duly rated by a human grader.

Four different essay questions, all rated on a scale of 5, were provided. Significantly, along with the essays, equal emphasis was given to collecting the instructions provided to the students. Thus, a full-fledged dataset consisting of 688 essays and their respective instructions, scores, and feedback were collected.

3.2 Data Preprocessing

The collected essays were preprocessed to prepare the textual data for machine learning. After manual copy-pasting into a text document, there was the removal of irrelevant characters and the discrepancies in the formatting. Common errors introduced during the copy-pasting were identified and corrected.

The preprocessed textual data was then converted into a structured JSON format. This conversion into JSON allowed the data to be organized in a format understandable by the model, thus allowing for smooth training and fine-tuning.

3.3 Model Fine-tuning and Training

The PEFT or the Parameter Efficient Fine-Tuning adapter layer plays a very crucial role in the model training process because the model is trained using it. This adaptation allows easy fine-tuning by the use of a limited set of parameters.

This was then succeeded by the incorporation of a specialized configuration that tailored the training process to realize optimal performance in essay grading. This configuration was also ideal for marking important parameters that determine the timeline of learning rates, regulation of batch size, and the application of regularization application techniques to curb overfitting. To finetune the Mistral 7b model optimally for use in an automated essay grading setting, the Supervised Fine-tuning method of SFT method was used also. This is used to reduce the coding requirement, extracting only the important and essential elements in the training of the model. Figure 2 shows a diagram of the finetuning process of Mistral 7b.

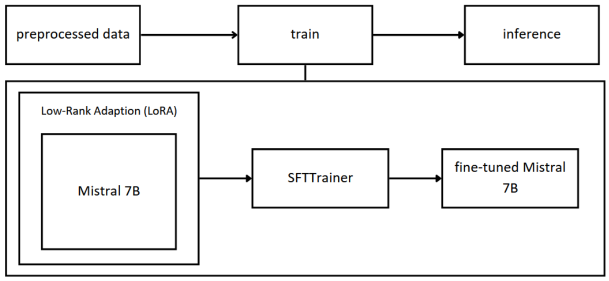

Figure 2: Diagram of the Training Process of Mistral 7b.

3.4 Model Performance Evaluation using BERT Score Metrics

BERT Score, the specially designed scoring metric for NLP, was applied in testing the trained and fine-tuned Mistral 7b model of this project. The semantic similarity between the machine-graded score and the human-graded score is checked. What is more is that the BERT Score differs from old-fashioned n-gram-based metrics in that it uses some context information and can, therefore, give a more nuanced evaluation of the ability of the model to produce evaluations of essays that are semantically appropriate [17].

BERT Score comes with three different evaluation criteria: precision, recall, and F1-score; they give finer details regarding how the model is performing in terms of accuracy while finding the correct grades. Precision reflects the percentage of true positive in all positive predictions. Recall reflects the proportion of actual high-score essays. F1-score reflects the balanced average of these quantities in the calling of each class. For a given the reference x, representing the expected output and a candidate $\hat{x}$, representing the actual output.

$$\mathrm{R_{BERT}} = \frac{1}{|\mathrm{x}|}\sum_{\mathrm{x_i}\in\mathrm{x}} \max_{\widehat{\mathrm{x_j}}\in\hat{\mathrm{x}}} \mathrm{x_i^T}\widehat{\mathrm{x_j}},\ \ \mathrm{P_{BERT}} = \frac{1}{|\hat{\mathrm{x}}|}\sum_{\widehat{\mathrm{x_j}}\in\hat{\mathrm{x}}} \max_{\mathrm{x_i}\in\mathrm{x}} \mathrm{x_i^T}\widehat{\mathrm{x_j}},\ \ \mathrm{F_{BERT}} = 2\frac{\mathrm{P_{BERT}}\cdot\mathrm{R_{BERT}}}{\mathrm{P_{BERT}}+\mathrm{R_{BERT}}} \tag{1}$$

For that, each full score was processed by determining the aligned token of the reference sequence x for each token of the candidate sequence x̂, which is corrected so that the recall is removed. On the other hand, removing the precision was performed by aligning in the other sentence to remove the alignment of each token in the reference to the most similar token. Then, the F1 measure was calculated by collating the precision and the recall [17]. The computed scores have the same numerical range of cosine similarity (between -1 and 1). The calculated scores lie within the numerical scale of the cosine similarity; however, to obtain an interpretable score, BERTScore was scaled based on an empirically determined lower bound. A baseline b was used in this regard. In this manner, the numerical values of the calculated scores actually fall within 0-1 themselves. In no way did ranking using BERTScore change next. But its job is only to make the score readable. On the sake of completeness, given below is the formula of calculated scores of $\hat{R}_{\mathrm{BERT}}$ of R_{BERT}, $\hat{P}_{\mathrm{BERT}}$ of P_{BERT},and $\hat{F}_{\mathrm{BERT}}$ of F_{BERT}.

$$\hat{R}_{\mathrm{BERT}} = \frac{R_{\mathrm{BERT}} - b}{1-b},\ \ \hat{P}_{\mathrm{BERT}} = \frac{P_{\mathrm{BERT}} - b}{1-b},\ \ \hat{F}_{\mathrm{BERT}} = \frac{F_{\mathrm{BERT}} - b}{1-b} \tag{2}$$

3.5 Moodle Plugin Development

A custom Moodle plugin was developed that seamlessly integrated the fine-tuned Mistral 7b model into the Moodle Platform, revolutionizing the process of essay evaluation. This plugin integrates the capability of Google Vision OCR technology, whereby the text is automatically extracted from the submitted image of handwritten essays. After submission, graders will then be interacting with the familiar Moodle interface, which initiates the analysis process of the model. Google Vision OCR will convert handwritten text within images to machine-readable text, thus providing the ground for the OCR component of the plugin.

Later on, the text data undergoes intensive processing by the fine-tuned and trained model of Mistral 7b. By using natural language processing techniques, Mistral 7b analyzes the essays' content. At the end of the evaluation, the plug-in produces an output for evaluation. This output contains the predicted grades and, if applicable, individual feedback regarding each essay. It is presented to them through a modal display format to ensure ease of accessibility and user friendliness. The grader is then able to review and make use of this report for the finalization of the grade and feedback of the essay.

This plug-in enables instructors with an effective and customizable essay evaluation instrument, bringing optimization to the grading process and to the general learning process within the Moodle environment. This report displays the assessment of the essay for instructors to review and, if desired, adjust. This plug-in empowers instructors with an efficient and customizable tool for essay evaluation.

3.6 User Perception Metrics for Moodle Plugin

User perception metrics play an important role in assessing the effectiveness of the Moodle plugin developed in this project for automated essay evaluation. The subsequent metrics were adopted to evaluate the usability, comprehensibility, and relevance of the plugin among the faculty members at CIT University. Five instructors were chosen to evaluate the Moodle Plugin:

3.6.1 Usability. This area dealt with the assessment of the user-friendliness and navigability of the Moodle plugin for educators. It encompassed how the introduction process, configuration settings, navigation through the different functionalities, clarity of instructions, and the general user experience were clear, including the installation, setup, and utilization.

- **Q1 -** The introduction process to the Moodle plugin was straightforward and simple to comprehend.
- **Q2 -** The configuration settings within the Moodle plugin were clear and intuitive.
- **Q3 -** Navigating through the different functionalities and sections of the Moodle plugin was smooth and easy.
- **Q4 -** The instructions provided for using the Moodle plugin were clear and straightforward.
- **Q5 -** Overall, the usability of the Moodle plugin, including installation, setup, and utilization, met my expectations.

3.6.2 Comprehensibility. Comprehensibility was judged based on how easily the educator could interpret and understand the generated essay scores and feedback provided by the Moodle plugin. It

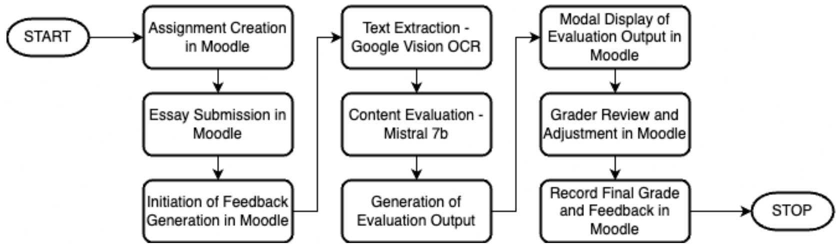

Figure 3: Essay Scorer Feedback Moodle plugin system process workflow diagram.

valued the accuracy and precision of the generated scores in respect to the learning objective, clarity of prompts, comprehensibility of instructions, and general effectiveness in facilitating student learning.

- **Q1 -** The automatically generated essay scores and feedback within the Moodle plugin were precise and accurate.
- **Q2 -** The essay scoring and feedback feature of the plugin aligned well with the instructions and intended learning objectives.
- **Q3 -** The generated scores and feedback provided accurate and concise prompts for grading.
- **Q4 -** The generated scores and feedback provided are accurate and aligned with the intended feedback based on the instructions.
- **Q5 -** Overall, the comprehensibility of the generated score and feedback within the Moodle plugin was satisfactory and contributed effectively to the learning experience.

3.6.3 Relevance. Relevance was examined to find the extent to which the generated scores and feedback within the Moodle plugin were relevant to the educational content, learning outcomes, instructional context, and intended audience. It was assessed whether the feedback provided was meaningful, applicable, and contributed meaningfully to the educational experience for the support of student achievement and academic success.

- **Q1 -** The generated scores and feedback within the Moodle plugin were relevant to the specific educational content or topic areas.
- **Q2 -** The scores and feedback generated by the plugin effectively addressed the desired learning outcomes or objectives.
- **Q3 -** The generated scores and feedback aligned well with the context in which they were instructed.
- **Q4 -** The score and feedback generation functionality of the Moodle plugin provided feedback suitable for the intended audience.
- **Q5 -** Overall, the relevance of the generated questions within the Moodle plugin was high and contributed significantly to the educational experience.

4 Result and Discussion

4.1 Model Performance Evaluation using BERT Score Metrics Results

A 10% sample of the collected essays was used to evaluate the performance of the fine-tuned and trained Mistral 7b model. Figure 4 below shows the comparison between the model-predicted scores with human-assigned scores. The model-predicted scores were generally aligned with actual scores as the two sets of scores showed a close parallel, can be seen in the figure. However, there was a significant outlier found within the data set. This likely represented an essay or a small set of essays in the test set where the model's grade was considerably far from the grades assigned by human graders.

Table 1: Average, Min and, Max BERT Scores

	Precision	Recall	F1
AVERAGE	88.41%	83.33%	85.78%
MAX	95%	90%	91%
MIN	86%	79%	82%

According to the BERT Score analysis, the precision, recall, and F1 scores were observed to be more or less consistent over all the essays. This is illustrated in Figure 5, indicating the stability in the performance of the model during the assessment process.

Further BERT Score analysis shown in Table 1, revealed consistent performance with a precision of 88.41% (min: 86%, max: 95%), indicating the model was able to correctly identify highly scoring essays. In addition, recall at 83.33% (min: 79%, max: 90%) indicates the model was able to capture a large number of compelling essays previously identified by human graders. Finally, the F1-score of 85.78% (min: 82%, max: 91%) demonstrates the model's overall stability in replicating human-grading decisions with a high degree of consistency.

4.2 Evaluation of Moodle Plugin using User Perception Metrics Results

Table 2 below highlights the results of the external testing where user perception of the usability, comprehensibility, and relevance of the system is highlighted.

The average of 4.28 in the usability score suggests that there is a solid foundation for that. A greater proportion of users found the system easy to use and navigate; that implies, to a great extent, the system has met the expectations of the users and provides excellent support for interaction. The average comprehensibility of 3.68 suggests moderate comprehensibility. Even though lower than the usability score, at 3.68 out of 5, it still reflects users' understanding of the features and instructions of the system. However, there could have been some areas where further enhancements could

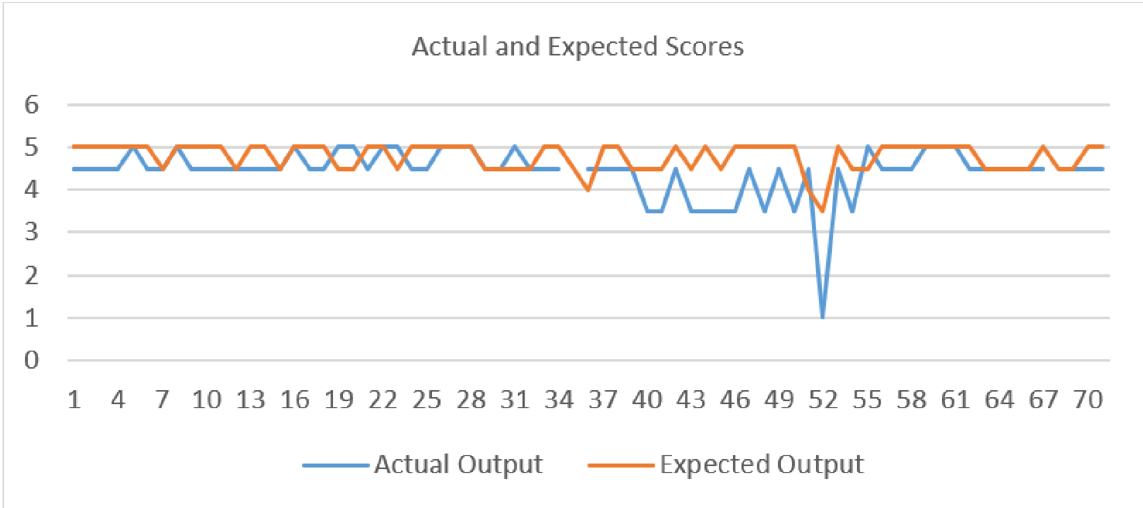

Figure 4: Actual Score (Model-predicted) and Expected Score (Human-assigned).

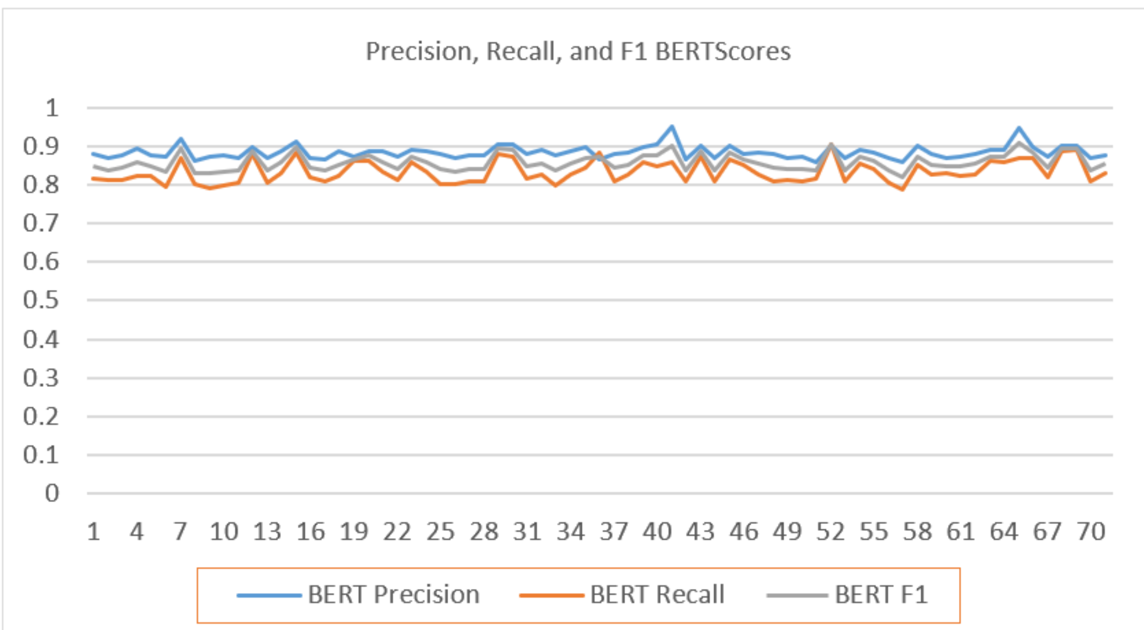

Figure 5: Comparison of the Precision, Recall, and F1 BERT Scores.

Table 2: Result of External Testing per User Perception Metrics

Q#	Metric Score (5 – Highest, 1 – Lowest)		
	Usability	Comprehensibility	Relevance
Q1	4.4	3.8	4
Q2	4.2	3.6	3.8
Q3	4.2	3.8	4.2
Q4	4.4	3.6	4
Q5	4.2	3.6	4
Average	4.28 (85.6%)	3.68 (73.6%)	4 (80%)

improve comprehensibility even more. The average score of 4.0 implies prominent relevance. It means a great proportion of users found the system's content and features relevant to their needs and expectations. It can be said from the results that the system has been addressing the users' requirements.

From the average scores in all three categories, the system is well-received. The very positive results in terms of usability and relevance indicate a user-friendly and valuable tool. However, in comprehensibility, the score being somewhat lower than the other two hints at more potential to improve user understanding with more clearly stated explanations and, in some cases, redesigned interfaces.

5 Conclusion and Recommendations

This study investigated the adoption of AI towards solving the problem of time-consuming and potentially subjective manual essay grading. Using OCR and a fine-tuned and trained LLM, Mistral 7b, an automated essay evaluation Moodle plugin system was developed. The integrated Moodle plugin system seems very promising for replicating human grading decisions and style, as evidenced by the close alignment between the model's predictions and human-assigned scores. Further, the analysis of BERT Score strengthens this conclusion by pointing out consistent precision, recall, and F1-score across the essays, giving an indication of the model's ability to produce semantically accurate evaluations.

User perception metrics add further insight into the system's performance. In fact, usability scores show that the system is useable because the user can interact with it effectively and intuitively. The comprehensibility scores suggest it is understandable, but there is still room for improvement. Relevance scores suggest that the users find the system's features and content acceptable, meeting their needs and expectations. These positive scores on user perception would suggest that educators are likely to find the system beneficial for daily tasks, hence practical value in educational settings.

The key recommendation, though this study demonstrates the feasibility of AI-powered essay evaluation, is further improvement. The model's generalizability and ability to handle a wider range of essay topics and writing styles can be improved by increasing the size of the dataset used for training. On the other hand, the current dataset focused only on one course, "IT031 - Living in the IT Era", and expanding the dataset to different educational courses would improve the generalizability of the system. The evaluation metrics employed in this study primarily revolve around semantic similarity between human-assigned and model-generated grades and feedback. Future research can add more measures to assess the model's performance in evaluating specific aspects of the quality of writing, which could include: cohesion, syntax, vocabulary usage, usage of phrases, grammar, and observance of conventions in writing. By adding this holistic parameter, it becomes possible to evaluate the model's capacity.

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